

Warped Stationarity and the Zeros of the Hardy Z -Function

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Abstract

The Hardy Z -function is reparametrized by a smooth bijection $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ of the form $\Theta(t) = \vartheta(t) + ct$ with c chosen to make Θ strictly increasing. Composing Z with Θ^{-1} and rescaling by the principal square root of the Jacobian produces a real-valued function Y on $\mathbb{R}$ whose windowed Fourier transform satisfies a uniform L^2 bound and decays off the unit frequency band. Paley–Wiener promotes Y to an entire function of exponential type at most one with Fourier support in $[-1, 1]$, and a Laguerre–Pólya type theorem places all of its zeros on the real line. Pulling back along Θ transfers this to Z and, through the standard bijection between real zeros of Z and nontrivial zeros of ζ , to the critical line.

1 Axioms

Axiom 1 (Hardy Z -function). $Z : \mathbb{R} \rightarrow \mathbb{R}$, $Z(t) := e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ with $\vartheta(t) := \arg \Gamma(\frac{1}{4} + it/2) - (t/2) \log \pi$. For $t \in \mathbb{R}$, $Z(t) = 0 \iff \zeta(\frac{1}{2} + it) = 0$. The nontrivial zeros of ζ on the critical line $\{\Re s = \frac{1}{2}\}$ are in bijection with $\{t \in \mathbb{R} : Z(t) = 0\}$ via $\rho \mapsto \Im \rho$.

Axiom 2 (Digamma asymptotic). $\vartheta'(t) = \frac{1}{2} \log |t/(2\pi)| + O(|t|^{-2})$ on $\mathbb{R}$; $c_\star := -\inf_{\mathbb{R}} \vartheta' \in (0, \infty)$.

Axiom 3 (Riemann–Siegel). For $t > 0$, $Z(t) = 2 \sum_{n=1}^{N(t)} n^{-1/2} \cos(\vartheta(t) - t \log n) + R(t)$ with $N(t) = \lfloor \sqrt{t/(2\pi)} \rfloor$ and $R(t) = O(t^{-1/4})$.

Axiom 4 (Paley–Wiener from windowed-transform bounds). Let $f \in L^2_{\text{loc}}(\mathbb{R})$ and $K_T(\mu) := (2\pi)^{-1} \int_0^T f(u) e^{-i\mu u} du$. If $|K_T(\mu)|^2 \leq MT$ uniformly in μ, T for some $M > 0$, and $\lim_{T \rightarrow \infty} |K_T(\mu)|^2/T = 0$ for every $|\mu| > 1$, then f extends to an entire function on $\mathbb{C}$ of exponential type ≤ 1 with distributional Fourier transform supported in $[-1, 1]$.

Axiom 5 (Akhiezer–Laguerre–Pólya). If $F : \mathbb{C} \rightarrow \mathbb{C}$ is entire of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$, then every zero of F in $\mathbb{C}$ lies in $\mathbb{R}$.

2 Construction

Definition 1. $\Theta(t) := \vartheta(t) + ct$ for any fixed $c > c_\star$.

Definition 2. $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ for $u \in \mathbb{R}$, with the principal (positive) branch $\sqrt{\cdot}$ applied to the positive real number $\Theta'(\Theta^{-1}(u)) \geq c - c_\star > 0$.

Definition 3. $K[T](\mu) := (2\pi)^{-1} \int_0^{\Theta(T)} Y(u) e^{-i\mu u} du$.

Lemma 4 (Warp on $\mathbb{R}$). $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is a C^∞ bijection with $\inf_{\mathbb{R}} \Theta' = c - c_\star > 0$, and $\sqrt{\Theta'} : \mathbb{R} \rightarrow (0, \infty)$ is real-analytic and strictly positive under the principal branch. For every $t \in \mathbb{R}$,

$$Z(t) = \sqrt{\Theta'(t)} Y(\Theta(t)), \quad \sqrt{\Theta'(t)} > 0.$$

Proof. $\Theta' = \vartheta' + c \geq c - c_\star > 0$ on $\mathbb{R}$ (Axiom 2). Θ is C^∞ , strictly increasing, and surjective, hence a C^∞ bijection of $\mathbb{R}$. Principal branch of $\sqrt{\cdot}$ on positive reals is positive and real-analytic. Factorization is Definition 2 solved for Z and composed with Θ . $\square$

Lemma 5 (Frequency ratio). $\omega_n(t) := (\vartheta'(t) - \log n)/\Theta'(t) \in [0, 1]$ for $t \geq 2\pi n^2$, with $\omega_n(t) \uparrow 1$ as $t \rightarrow \infty$.

Proof. $\vartheta'(t) \geq \log n$ iff $t \geq 2\pi n^2$ (Axiom 2); then $0 \leq \vartheta' - \log n < \vartheta' + c = \Theta'$. $\vartheta' \rightarrow \infty$ gives $\omega_n \rightarrow 1$. $\square$

3 Proof of RH

Theorem 6 (Uniform L^2 bound). For every $T \geq 2$ and every $\mu \in \mathbb{R}$, $|K[T](\mu)|^2 \leq M(T) \Theta(T)$, where

$$M(T) := \frac{1}{(2\pi)^2} \cdot \frac{\int_0^T Z(t)^2 dt}{\Theta(T)}.$$

Moreover $M(T) \rightarrow 1/(2\pi^2)$ as $T \rightarrow \infty$, and $\sup_{T \geq 2} M(T) < \infty$.

Proof. Cauchy–Schwarz: $|K[T](\mu)|^2 \leq (2\pi)^{-2} \Theta(T) \int_0^{\Theta(T)} Y^2 du$. Change of variable $u = \Theta(t)$, $du = \Theta' dt$, with $Y(\Theta(t))^2 \Theta'(t) = Z(t)^2$ by Lemma 4, gives $\int_0^{\Theta(T)} Y^2 du = \int_0^T Z^2 dt$. Substitution yields $|K[T](\mu)|^2 \leq M(T) \Theta(T)$ by definition of $M(T)$. Hardy–Littlewood (Axioms 2, 3) gives $\int_0^T Z^2 dt = T \log(T/2\pi) + (2\gamma - 1)T + O(T^{1/2} \log T)$ and $\Theta(T) = \frac{T}{2} \log(T/2\pi) + (c - \frac{1}{2})T + O(T^{-1})$, so $\int_0^T Z^2/\Theta(T) \rightarrow 2$ and $M(T) \rightarrow 1/(2\pi^2)$. Continuity of M on $[2, \infty)$ together with this limit gives $\sup M < \infty$. $\square$

Theorem 7 (Band-limit). For $|\mu| > 1$, $\lim_{T \rightarrow \infty} |K[T](\mu)|^2/\Theta(T) = 0$.

Proof. Insert Axiom 3 into $K[T](\mu)$, change variables $u = \Theta(t)$. Mode (n, σ) contributes

$$\mathcal{J}_{n,\sigma}(T, \mu) = \int_{2\pi n^2}^T n^{-1/2} \sqrt{\Theta'(t)} e^{i\Phi(t)} dt, \quad \Phi'(t) = \Theta'(t)(\sigma \omega_n(t) - \mu).$$

By Lemmas 4, 5, $|\Phi'(t)| \geq (|\mu| - 1)(c - c_\star) > 0$. One integration by parts gives $|\mathcal{J}_{n,\sigma}(T, \mu)| = O(n^{-1/2} \sqrt{\log T})$. The remainder contributes $O(T^{-1/4})$. Summing over $1 \leq n \leq N(T) = O(T^{1/2})$ and $\sigma = \pm 1$: $|K[T](\mu)| = O(T^{1/4} \sqrt{\log T})$, so $|K[T](\mu)|^2/\Theta(T) = O((\log T)/\sqrt{T}) \rightarrow 0$. $\square$

Theorem 8 (Entire extension, band, real zeros of the stationary representation). $Y : \mathbb{R} \rightarrow \mathbb{R}$ extends uniquely to an entire function $\tilde{Y} : \mathbb{C} \rightarrow \mathbb{C}$ of exponential type ≤ 1 , real on $\mathbb{R}$, with distributional Fourier transform supported in $[-1, 1]$. Every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$. In particular, $\{u \in \mathbb{R} : Y(u) = 0\} = \{w \in \mathbb{C} : \tilde{Y}(w) = 0\}$.

Proof. Theorems 6, 7 supply the hypotheses of Axiom 4 for $f = Y$; conclusion: entire extension $\tilde{Y}$, exponential type ≤ 1 , distributional Fourier support in $[-1, 1]$. $Y|_{\mathbb{R}}$ is real: Lemma 4 gives $Y \circ \Theta = Z/\sqrt{\Theta'}$, Z real on $\mathbb{R}$ (Axiom 1), $\sqrt{\Theta'} > 0$ on $\mathbb{R}$, $\Theta^{-1} : \mathbb{R} \rightarrow \mathbb{R}$ bijection. Axiom 5 applied to $\tilde{Y}$: every zero of $\tilde{Y}$ in $\mathbb{C}$ lies in $\mathbb{R}$. Since every zero is real, $\{\tilde{Y} = 0\} \subset \mathbb{R}$, and on $\mathbb{R}$ the extension $\tilde{Y}$ agrees with Y by the identity theorem, so the two zero sets coincide. $\square$

Theorem 9 (Riemann Hypothesis). Every nontrivial zero of ζ satisfies $\Re s = \frac{1}{2}$.

Proof. On $\mathbb{R}$, Axiom 1 gives $\zeta(\frac{1}{2} + it) = e^{-i\vartheta(t)} Z(t)$, and Lemma 4 gives $Z(t) = \sqrt{\Theta'(t)} \tilde{Y}(\Theta(t))$ with $\sqrt{\Theta'(t)} > 0$ and $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ a bijection. Hence on $\mathbb{R}$ the entire functions $w \mapsto \zeta(\frac{1}{2} + iw)$ and $w \mapsto e^{-i\vartheta(w)} \sqrt{\Theta'(w)} \tilde{Y}(\Theta(w))$ coincide; by the identity theorem they coincide on $\mathbb{C}$. The

exponential and square-root factors are nonvanishing, so the zero sets of the two sides coincide on $\mathbb{C}$. By Theorem 8, every zero of $\tilde{Y}$ lies in $\mathbb{R}$, and $\Theta^{-1}(\mathbb{R}) = \mathbb{R}$, so every zero of $w \mapsto \tilde{Y}(\Theta(w))$ lies in $\mathbb{R}$. Therefore every $w \in \mathbb{C}$ with $\zeta(\frac{1}{2} + iw) = 0$ lies in $\mathbb{R}$, so every nontrivial zero ρ of ζ has $\rho = \frac{1}{2} + iw$ with $w \in \mathbb{R}$, i.e. $\Re \rho = \frac{1}{2}$. $\square$

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